Tutorial 1

Matthew Farah
June 19, 2026

Contents

Problem 1:
Define a binary operator $*$ on $\mathbb{Z} \times \mathbb{Z}$ by:

$$(a, b) * (c, d) \coloneqq (ab + bc, bd)$$

Is $\mathbb{Z} \times \mathbb{Z}$ a group under this operation? If so, prove it by establishing which group axioms fail (with proofs/counterexamples).

Solution:

$\mathbb{Z} \times \mathbb{Z}$ does <u>not</u> constitute a group under $*$. To see why, we'll examine which of the group axioms fail to hold and why. As a refresher, for $(\mathbb{Z} \times \mathbb{Z}, *)$ to be a group, it must satisfy all of the following four criteria:

1. $*$ must be a binary operator on $\mathbb{Z} \times \mathbb{Z}$.
2. $*$ must be associative over $\mathbb{Z} \times \mathbb{Z}$.
3. $*$'s identity element must be contained in $\mathbb{Z} \times \mathbb{Z}$.
4. Every element in $\mathbb{Z} \times \mathbb{Z}$ must have an inverse in $\mathbb{Z} \times \mathbb{Z}$ with respect to $*$.

We will investigate each criterion separately below.

(I) Criterion one.

The first criterion states that $*$ must be a binary operator on $\mathbb{Z} \times \mathbb{Z}$.

For $*$ to form a binary operator over $\mathbb{Z} \times \mathbb{Z}$, it must be defined and closer over all of $\mathbb{Z} \times \mathbb{Z}$. By inspection, and knowledge of the fact that $\mathbb{Z}$ is closed under both multiplication and addition, we can see that this is actually the case, and so $*$ does indeed constitute a valid binary operator over $\mathbb{Z} \times \mathbb{Z}$.

(II) Criterion two.

Criterion two states that $*$ must be associative over $\mathbb{Z} \times \mathbb{Z}$.

By counter example, we can see that this fails to hold true. Although $(1, 0), (0, 2)$, and $(3, 0)$ all belong to $\mathbb{Z} \times \mathbb{Z}$, we have that:

$$\begin{aligned} ((1, 0) * (0, 2)) * (3, 0) &= (2, 0) * (3, 0) \\ &= (0, 0) \end{aligned}$$

Whereas:

$$\begin{aligned} (1, 0) * ((0, 2) * (3, 0)) &= (1, 0) * (0, 6) \\ &= (6, 0) \end{aligned}$$

And thus, since $(a * b) * c \neq a * (b * c)$ for all $a, b, c \in (\mathbb{Z} \times \mathbb{Z})$, $*$ is not associative over its domain and so criterion two isn't satisfied.

(III) Criterion three.

Criterion three states that $*$ must have an identity element in $\mathbb{Z} \times \mathbb{Z}$.

Despite not being associative, it is actually the case that $*$ has an identity element in $\mathbb{Z} \times \mathbb{Z}$. To see this, let (a, b) be an arbitrary element in $\mathbb{Z} \times \mathbb{Z}$. Since:

$$\begin{aligned}
\boxed{(a,b) * (0,1)} &= (a(1) + b(0), b(1)) && \text{(By definition of } *) \\
&= \boxed{(a,b)} && \text{(Simplifying)} \\
&= (0(b) + 1(a), 1(b)) && \text{(Expanding)} \\
&= \boxed{(0,1) * (a,b)} && \text{(By definition of } *)
\end{aligned}$$

$(0, 1) \in (\mathbb{Z} \times \mathbb{Z})$ therefore satisfies $(a, b) * (0, 1) = (a, b) = (0, 1) * (a, b)$ for arbitrary $(a, b) \in \mathbb{Z} \times \mathbb{Z}$, making it an identity element for $*$ in its domain. Thus, $*$ really does have an identity element in $\mathbb{Z} \times \mathbb{Z}$, satisfying criterion three.

(IV) Criterion four.

Lastly, criterion four requires that every element in $\mathbb{Z} \times \mathbb{Z}$ have an inverse under $*$ that's contained in $\mathbb{Z} \times \mathbb{Z}$.

$*$ fails to satisfy criterion four as well. Taking $(1, 0)$ as an example, we can see it to have an inverse in $\mathbb{Z} \times \mathbb{Z}$ there must exist some $(a, b) \in (\mathbb{Z} \times \mathbb{Z})$ such that:

$$(1,0) * (a,b) = (0,1) \qquad \text{(By definition of inverse)}$$

Since we determined as a part of our analysis of criterion three that $(0, 1)$ must be the identity of $*$ over $\mathbb{Z} \times \mathbb{Z}$. However, since:

$$\begin{aligned}
(1,0) * (a,b) &= (0,1) && \text{(By our earlier equality)} \\
\implies ((1)(b) + (0)(a), (0)(b)) &= (0,1) && \text{(By definition of } *) \\
\implies (b, 0) &= (0,1) && \text{(Simplifying)} \\
\implies 0 &= 1 && \text{(Equating entries)}
\end{aligned}$$

Since this clearly cannot be the case, $(1, 0)$ (and actually, any element of the form $(x, 0)$) cannot have an inverse in $\mathbb{Z} \times \mathbb{Z}$ under $*$. Hence, criterion four is violated as well.

Problem 2:
Let G be a group and let $a, b \in G$. Prove that the inverse of ab is $b^{-1}a^{-1}$.

Solution:

Let $a, b \in G$ be arbitrary and let $a^{-1}, b^{-1} \in G$ be their respective inverses. To show that the inverse of ab is $b^{-1}a^{-1}$, we need to show that $(ab)(b^{-1}a^{-1}) = (b^{-1}a^{-1})(ab) = 1$. Thankfully, this can be proven directly and relatively easily using the following manipulations:

$$\begin{aligned}
\boxed{(ab)(b^{-1}a^{-1})} &= a(b(b^{-1}a^{-1})) && \text{(By associativity)} \\
&= a((bb^{-1})a^{-1}) && \text{(By associativity)} \\
&= a((1)a^{-1}) && \text{(By definition of inverse)} \\
&= aa^{-1} && \text{(By definition of identity)} \\
&= \boxed{1} && \text{(By definition of inverse)} \\
&= b^{-1}b && \text{(By definition of inverse)} \\
&= b^{-1}((1)b) && \text{(By definition of identity)} \\
&= b^{-1}((a^{-1}a)b) && \text{(By definition of inverse)} \\
&= (b^{-1}(a^{-1}a))b && \text{(By associativity)} \\
&= ((b^{-1}a^{-1})a)b && \text{(By associativity)} \\
&= \boxed{(b^{-1}a^{-1})(ab)} && \text{(By associativity)}
\end{aligned}$$

And so, the inverse of ab is indeed $b^{-1}a^{-1}$!

Problem 3:
Let G be a group and let $a, b \in G$. Prove that if a and b commute, then so do a^{-1} and b^{-1}.

Solution:

Once again, let $a, b \in G$ be arbitrary elements such that a and b commute, and let $a^{-1}, b^{-1} \in G$ be their respective inverses. To show that a^{-1} and b^{-1} commute, we must prove that $a^{-1}b^{-1} = b^{-1}a^{-1}$. Since a and b are assumed to be commutative, we know that $ab = ba$. Using this, we have that:

$$\begin{aligned}
a^{-1}b^{-1} &= 1(a^{-1}b^{-1}) && \text{(By definition of identity)} \\
&= ((b^{-1}a^{-1})(ab))(a^{-1}b^{-1}) && \text{(By definition of inverse)} \\
&= (b^{-1}a^{-1})((ab)(a^{-1}b^{-1})) && \text{(By associativity)} \\
&= (b^{-1}a^{-1})((ba)(a^{-1}b^{-1})) && \text{(By commutativity of } a \text{ and } b\text{)} \\
&= (b^{-1}a^{-1})(1) && \text{(By definition of inverse)} \\
&= b^{-1}a^{-1} && \text{(By definition of identity)}
\end{aligned}$$

And so, since $a^{-1}b^{-1} = b^{-1}a^{-1}$, we indeed have that a^{-1} and b^{-1} commute.

Problem 4:
Prove that a group with exactly four elements must be abelian.

(*Hint*: Let $G := \{ 1, a, b, c \}$ and make a multiplication table for G by brute force. Start by showing either that $ab = 1$ or $ab = c$; thus proceed by brute force.)

Solution: